SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4**	**Assessment:**	**Assessment Tool 3**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	**Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)**		
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Section / Batch:	**CSE-AI**	**Date:**	**24/08/2026**

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____G.VEDAVIKAS GOWD_____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, wellstructured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

Introduction to Statistical Inference

1. **What is statistical inference?**

Answer: Statistical inference is the process of using sample data to make conclusions about a population. It helps estimate unknown population values and test different claims. It is commonly used in statistics and data science. It supports decision-making based on available data.

2. **Why is statistical inference important in data science?**

Answer: Statistical inference helps data scientists make decisions using sample data. It allows them to make predictions and conclusions about a larger population. It is useful for estimating unknown values and testing assumptions. Therefore, it supports reliable datadriven decisions.

3. **What is the difference between descriptive statistics and inferential statistics? Answer:** Descriptive statistics summarizes and describes collected data using measures such as mean and median. Inferential statistics uses sample data to make conclusions about a

population. Descriptive statistics explains the available data. Inferential statistics helps generalize results to a population.

4. **What is a population in statistics?**

Answer: A population is the complete group of individuals, objects, or observations being studied. It may include people, products, customers, or measurements. For example, all students in a college can form a population. Statistical analysis is used to understand the characteristics of the population.

5. **What is a sample?**

Answer: A sample is a smaller group selected from a population for statistical analysis. It is used when studying the complete population is difficult or expensive. A good sample should represent the population properly. Sample results can be used to make conclusions about the population.

6. **Why do data scientists usually work with samples instead of complete populations?** **Answer:** Studying the entire population can require a lot of time, money, and resources. In some cases, collecting information from everyone is not possible. A properly selected sample can provide useful information about the population. Therefore, samples are commonly used for statistical analysis.

7. **What is a population parameter?**

Answer: A population parameter is a numerical value that describes a characteristic of an entire population. Examples include population mean, variance, and proportion. Usually, the parameter is unknown. It can be estimated using sample data.

8. **What is a sample statistic?**

Answer: A sample statistic is a numerical value calculated from sample data. Examples include sample mean, sample variance, and sample proportion. It is used to estimate a population parameter. Its value may change when a different sample is selected.

9. **Give a real-world example where statistical inference is used in industry. Answer:** A company can survey a sample of customers to measure their satisfaction. The sample results can be used to estimate the satisfaction level of all customers. The company does not need to survey every customer. This is an example of statistical inference.

10. **What are the two major activities involved in statistical inference?**

Answer: The two major activities are estimation and hypothesis testing. Estimation is used to estimate unknown population parameters. Hypothesis testing is used to test claims about a population. Both activities use sample data

B. Frequentist Approach

11. What is the frequentist approach to statistical inference?

Answer: The frequentist approach is based on sample data, repeated sampling, and long-run probabilities. It considers population parameters as fixed but unknown values. The sample data is considered random. It is commonly used in confidence intervals and hypothesis testing.

12. What is the basic idea behind frequentist statistics?

Answer: Frequentist statistics uses repeated sampling to make statistical conclusions. It focuses on how often an event occurs in the long run. The population parameter is treated as fixed. Sample data is used to estimate or test the parameter.

13. How does the frequentist approach differ from the Bayesian approach?

Answer: The frequentist approach treats population parameters as fixed but unknown. The Bayesian approach treats parameters as uncertain. Bayesian methods can also use prior information. Frequentist methods mainly use sample data and repeated sampling.

14. In the frequentist approach, what is considered fixed: the parameter or the data?

Answer: In the frequentist approach, the population parameter is considered fixed but unknown. The sample data is considered random. Different samples can produce different results. Therefore, the randomness comes from the sampling process.

15. What does long-run frequency mean in frequentist statistics?

Answer: Long-run frequency means the proportion of times an event occurs when an experiment is repeated many times. For example, repeated coin tossing can be used to observe the frequency of heads. This frequency approaches a stable value. It is an important idea in frequentist statistics.

16. Why is random sampling important in the frequentist approach?

Answer: Random sampling gives members of the population a fair chance of being selected. It reduces selection bias and helps the sample represent the population. It also supports probability calculations. Therefore, it helps produce reliable statistical conclusions.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?

Answer: No, a population parameter is not considered a random variable in the frequentist approach. It is treated as a fixed but unknown value. The sample data changes from one sample to another. Therefore, randomness comes from the data rather than the parameter.

18. Give an example of a frequentist problem in a data science application. Answer: A company may introduce a new advertisement and want to know whether it increases the conversion rate. A sample of visitors can be collected and analyzed. A hypothesis test can be performed using the sample data. This is an example of a frequentist problem.

C. Point Estimation

19. What is a point estimate?
Answer: A point estimate is a single numerical value used to estimate an unknown population parameter. For example, the sample mean can estimate the population mean. It provides one best estimated value. However, it does not show the uncertainty of the estimate.

20. What is an estimator?
Answer: An estimator is a statistical formula or rule used to estimate a population parameter. For example, the sample mean is an estimator of the population mean. It is calculated using sample data. The numerical result obtained is called the estimate.

21. What is the difference between an estimator and an estimate?
Answer: An estimator is the formula or method used to estimate a parameter. An estimate is the actual numerical value obtained from the sample. For example, $\bar{x}$ is an estimator of the population mean. If $\bar{x}$ is 50, then 50 is the estimate.

22. What is the point estimator for the population mean?
Answer: The sample mean is the point estimator for the population mean. It is represented by $\bar{\mathbf{x}}$. The formula is $\bar{\mathbf{x}} = \boldsymbol{\Sigma}\mathbf{x}/\mathbf{n}$. It is widely used because it is an unbiased estimator of the population mean.

23. How do you estimate a population proportion from a sample?
Answer: A population proportion is estimated using the sample proportion. It is represented by $\hat{\mathbf{p}}$. The formula is $\hat{\mathbf{p}} = \mathbf{x}/\mathbf{n}$, where x is the number of successes and n is the sample size. It gives an estimate of the population proportion.

24. What is the point estimate of population variance?
Answer: Sample variance is commonly used as the point estimate of population variance. It is represented by $\mathbf{s^2}$. It measures the variation of observations around the sample mean. The usual sample variance is an unbiased estimator of population variance.

25. Why is the sample mean commonly used to estimate the population mean?

Answer: The sample mean is simple to calculate and uses all observations in the sample. It is an unbiased estimator of the population mean. Its accuracy generally improves as the sample size increases. Therefore, it is widely used in statistical inference.

26. What properties should a good estimator have?

Answer: A good estimator should ideally be unbiased, consistent, and efficient. An unbiased estimator gives the correct value on average. A consistent estimator becomes more accurate as the sample size increases. An efficient estimator has relatively low variance.

27. What is an unbiased estimator?

Answer: An unbiased estimator is an estimator whose expected value is equal to the true population parameter. It does not systematically overestimate or underestimate the parameter. The sample mean is an example of an unbiased estimator. It is an important property of a good estimator.

28. What is meant by the bias of an estimator?

Answer: Bias is the difference between the expected value of an estimator and the true population parameter. It shows systematic error in estimation. If the bias is zero, the estimator is unbiased. A lower bias generally indicates better accuracy.

29. What is the difference between bias and variance of an estimator?

Answer: Bias measures how far the expected estimate is from the true parameter. Variance measures how much the estimate changes between different samples. Low bias indicates accuracy on average. Low variance indicates consistency of the estimator.

30. Why can two different samples produce different point estimates for the same population parameter?

Answer: Different samples contain different observations from the same population. Therefore, their sample statistics can have different values. This difference is called sampling variation. It is a normal part of statistical inference.

D. Variability in Estimates

31. Why does a point estimate have uncertainty?

Answer: A point estimate is calculated from a sample rather than the entire population. Different samples can produce different estimates. Therefore, the exact population parameter may not be known. This uncertainty can be measured using standard error and confidence intervals.

32. What is the sampling distribution of an estimator?

Answer: The sampling distribution is the distribution of an estimator obtained from repeated samples. Each sample produces a different estimate. The collection of these estimates forms the sampling distribution. It helps us understand the variability of an estimator.

33. What is the standard error?

Answer: Standard error measures the variability of a sample statistic across repeated samples. It shows how precisely a statistic estimates a population parameter. A smaller standard error means greater precision. For a sample mean, it depends on population variation and sample size.

34. How is standard error different from standard deviation?

Answer: Standard deviation measures variation among individual observations in a dataset. Standard error measures variation of a sample statistic across repeated samples. Standard deviation describes the spread of data. Standard error describes the precision of an estimate.

35. How does increasing the sample size affect the standard error of the sample mean?

Answer: Increasing the sample size decreases the standard error. The formula is $\mathbf{SE = \sigma/\sqrt{n}}$. As the sample size increases, the denominator increases. Therefore, the sample mean becomes more precise.

36. Why does variability in an estimate decrease when the sample size increases? Answer:
A larger sample provides more information about the population. Individual random differences have less effect on the final estimate. Therefore, the estimate becomes more stable. This results in lower variability.

37. What is the relationship between standard error and confidence intervals? Answer:
Standard error affects the width of a confidence interval. A larger standard error produces a wider confidence interval. A smaller standard error produces a narrower interval.

Therefore, a lower standard error indicates greater estimation precision.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect?

Answer: No, it does not mean that one sample is incorrect. Different random samples can contain different observations. Therefore, their sample means can naturally be different. This difference is called sampling variation.

E. Confidence Intervals

39. What is a confidence interval?

Answer: A confidence interval is a range of values used to estimate an unknown population parameter. It is calculated from sample data at a selected confidence level. It provides more information than a single point estimate. It also shows the uncertainty of the estimate.

40. Why is a confidence interval more informative than a point estimate?

Answer: A point estimate provides only one numerical value. A confidence interval provides a range of reasonable values. It also indicates the uncertainty associated with the estimate. Therefore, it gives a more complete understanding of the population parameter.

41. What does a 95% confidence level mean in the frequentist approach?

Answer: A 95% confidence level means that if the same sampling procedure is repeated many times, about 95% of the resulting intervals will contain the true population parameter. It describes the long-run performance of the method. It does not mean a particular interval has a 95% probability of containing the parameter.

42. What is the difference between a confidence level and a confidence interval?

Answer: A confidence level is the percentage used to construct a confidence interval, such as 90%, 95%, or 99%. A confidence interval is the actual range calculated from sample data. For example, 95% is the confidence level. A range such as (40, 50) is the confidence interval.

43. What factors determine the width of a confidence interval?

Answer: The main factors are sample size, data variability, and confidence level. Higher variability makes the interval wider. A higher confidence level also makes the interval wider. A larger sample size generally makes the interval narrower.

44. How does increasing the sample size affect the width of a confidence interval?

Answer: Increasing the sample size generally makes the confidence interval narrower. It reduces the standard error of the estimate. This increases the precision of the result. Therefore, larger samples usually give more precise confidence intervals.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?

Answer: Increasing the confidence level from 95% to 99% makes the confidence interval wider. A higher confidence level requires a larger margin of error. Therefore, the range becomes larger. This provides greater confidence but lower precision.

46. What is the difference between a z-confidence interval and a t-confidence interval?

Answer: A z-confidence interval uses the standard normal distribution. A t-confidence interval uses the t-distribution. The t-distribution is commonly used when the population standard deviation is unknown. It is especially useful for smaller samples.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?

Answer: The t-distribution is used when the population standard deviation is unknown. The sample standard deviation is used instead. It is especially appropriate for small samples. It accounts for the additional uncertainty in estimating the standard deviation.

F. Hypothesis Testing Using Confidence Intervals

48. What is hypothesis testing?

Answer: Hypothesis testing is a statistical method used to test a claim about a population. It starts with a null hypothesis and an alternative hypothesis. Sample data is used to determine whether there is enough evidence against the null hypothesis. The result helps us make a statistical decision.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?

Answer: The null hypothesis, $\mathbf{H_0}$, represents the existing or default claim about a population. The alternative hypothesis, $\mathbf{H_1}$, represents the claim we want to find evidence for. Sample data is used to compare the two hypotheses. Based on the evidence, we reject or fail to reject H_0.

50. How can a confidence interval be used to make a decision about a hypothesis?

Answer: A confidence interval can be used to check whether the hypothesized parameter value lies inside the interval. If the value is outside the interval, we reject H_0 at the corresponding significance level. If it is inside the interval, we generally fail to reject H_0.

Thus, confidence intervals can be used for hypothesis-testing decisions.